How A Microprocessor Works

Here we illustrate a simple addition of two numbers, 3 and 4. The visual depicts functioning of a microprocessor during the addition process, irrespective of the data entry sequence

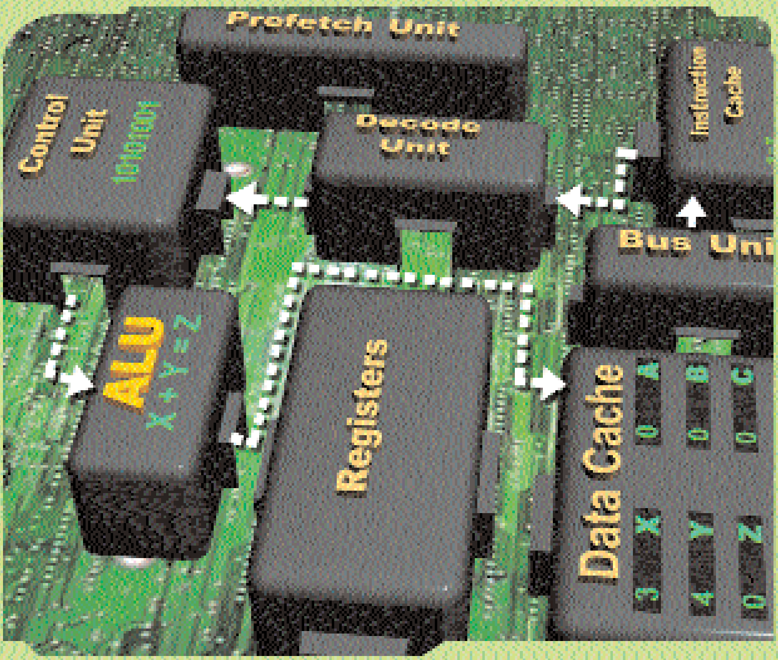
Step 3 : The '+' key is pressed

a. On pressing the '+' key, the µP is alerted, which asks the Prefetch Unit to get the instruction. The new data instruction comes into the µP through the Bus Unit, and is stored in the Instruction Cache as the code 'X + Y = Z', indicating that an addition operation will be performed.

b. The Prefetch Unit then sends a copy of this new data to the Decode Unit for processing.

c. At the Decode Unit, the data is translated into binary and sent to the Control Unit and Data Cache. Also, the ALU is alerted that the ADD function has to be performed.

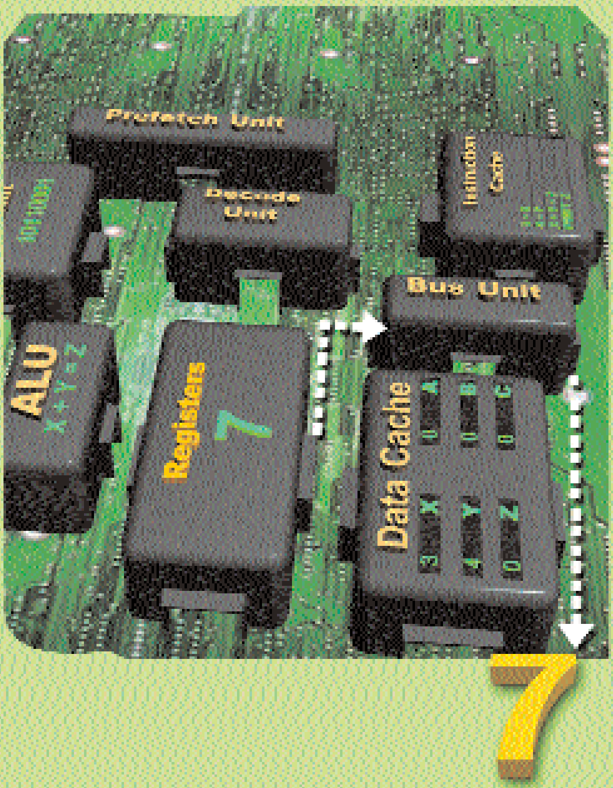
d. In the Control Unit, the code (X + Y = Z) is broken down, and the ADD command is sent to the ALU. In the ALU, the values of X and Y are added, and the result, '7', is sent to an address location in the Register.



Step 4 : The '=' key is pressed

a. On pressing the '=' key, the µP is alerted. It asks the Prefetch Unit to get the instruction. The new data instruction comes into the µP through the Bus Unit, and is stored in the Instruction Cache as 'Print Z'.

b. The Prefetch Unit then sends a copy of this new data to the Decode Unit for processing.



Imaging **Vikram Kathare, Shyam Shriram** and Text **Preethi Chamikutty**